Image Processing - Greenness

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December 5, 2018

1 What is greenness?

1.1 Introduction

Among the series studied over the years, a detection of greenish color patterns is one of the most investigated. In areas such as industry, agriculture, food quality, we can areas related to the detection of plants, herbs and diseases in plantations, control of the level of plant growth and appearance of green fruits and vegetables (pepper, pepper, cucumber), broadcasting of football and football on television and which use the coronal as the predominant characteristic.

1.2 Definition

By definition, greenness is the green intensity greater than blue or red intensities.

The Green Index is a more interesting index than the Vegetation Health Index is a larger range, is more sensitive, and the user has a more graduated scale for measurement.

Calculating a Greenness Index of the vegetation Healthy green vegetation absorbs visible light from the sun and reflects infrared light photosynthetic process. By exploring how the compensation properties of living vegetation can download a green vegetation index as a measure of the photosynthetic capacity of the surface cover terrestrial To determine the density of green on a piece of land, researchers should be distinct nuclei (wavelengths) of the visible light and the bubble.

1.3 Greenness index

In this article we try to implement the largest number of new indexes created for this article, along with the old ones for comparison. For the creation of new indexes, we use the existing formulas to detect other colors.

Source	Formula
Used modified of redness index [9]	$GreennesskG = kG - (R + B)$
Base on a redness index [2]	$Greenness_{min} = G - min(R, B)$
Based on Greenness index from 1991 [1]	$Greenness_{G+R} = (G + R)/(G - R)$
Greenness index from 1991 [1]	$Greenness_{G-R} = (G - R)/(G + R)$
Used modified of redness Smolka [4]	$Greenness_{Smolka} = (G - Max\{R, B\})^2/G$
Based on [3]	$Greenness_{G2} = (G^2)/(B^2 + R^2 + k)$
Text *Student Climate Data* [5]	$Greenness = (G)/(R + G + B)$
Base on I_{kaw} [6]	$I_{PCA} = 0.7582\|R - B\| - 0.1168\|R - G\| + 0.6414\|G - B\|$
A simpler method for detecting the color green [8]	0.16 - 0.50 $\rightarrow$ Hue 0.075 - 1.0 $\rightarrow$ Saturation 0.09 - 0.91 $\rightarrow$ Value
Most commonly fused strategy or Greenness identification [7]	$r = \frac{R}{R+G+B}$, $g = \frac{G}{R+G+B}$, $b = \frac{B}{R+G+B}$ $ExG = 2g - r - b$ $ExGR = Exg - (1.44 - g)$ $CIVE = 0.441r - 0.811g + 0.3855b + 18.78745$ $VEG = \frac{g}{r^{0.667}b^{0.333}}$ $COM = 0.25 * ExG + 0.30 * ExGR + 0.33 * CIVE + 0.12 * VEG$

2 Objective

The objectives of this project is to study and implement the greenness indices avaible in theliterature, and to numerically evaluate the segmentation power of each greenness index using a diversified base and appropriate metrics.

The implementation will occur in an existing software, however, the index is to identify browness, so my work will be reverse engineering to understand the implementation, and once you have mastery over the code, the implementation of the techniques will be done, substituting existing ones by the techniques of Greenness.

3 Results

The index that showed the best result was the smolka formula. In our tests the result was far more superior than some formulas. One of the biggest problems working with the detection of green is the acceptance of green, even if it is a light green or dark, clarity was a very recurrent point in our research, if a dark green should be considered as part of the plantation, or completely ignored. Some implemented indexes presented the binary image with the inverted colors,

others, ignored a green if it is very dark, or because it was for example marked by a vehicle. But regardless of the clarity, the Smolka was a pain results that got the best result, reaching an accuracy of up to 99%, becoming a surprise and a great tool for the detection of green. Below we compare the accuracy and precision of each of them, followed by images to compare the efficacy of the methods:

Index Used	**Average Precision**	**Average Accuracy**
Smolka	0.9996	0.9973
Min	0.9288	0.9560
KG	0.9716	0.9832
IPCA	0.3667	0.4337
Greenness	0.3018	0.2165
G+R	0.6540	0.6411
G-R	0.2428	0.6411

Figure 1: Imagem Original

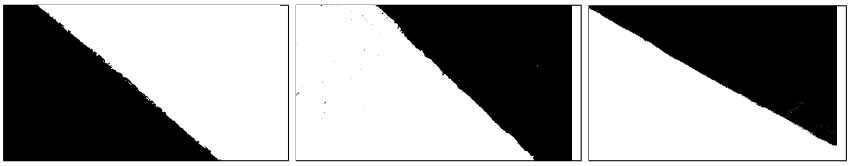

Figure 2: Imagem Smolka

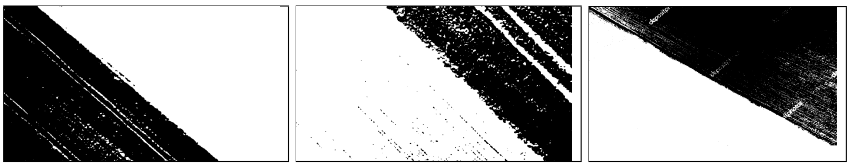

Figure 3: Imagem Min

Figure 4: Imagem KG

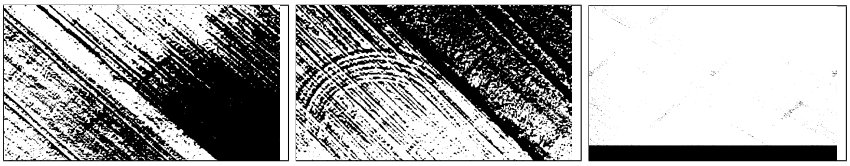

Figure 5: Imagem IPCA

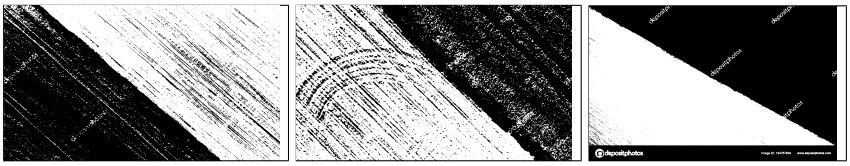

Figure 6: Imagem Greenness

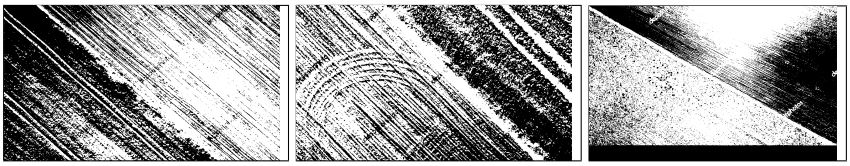

Figure 7: Imagem G+R

Figure 8: Imagem G-R

Figure 9: Imagem G2

References

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